

Sri Lanka Institute of Information Technology



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B.Sc. (Hons) in Information Technology

1. Title

Topic: Web-Based Restaurant and Event Management System

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3. Introduction

3.1 Project Overview

The **Web-Based Restaurant and Event Management System** is a comprehensive web application design to streamline and digitalize the operations of restaurants and event management services. Traditional restaurant and event management often rely on manual processing, and customer management. These manual methods can result in booking conflicts, inefficient resource utilization, increased operational costs, and reduce customer satisfaction.

This proposed system aims to integrate restaurant dining services and event management into a single centralized platform. Customers will be able to reserve restaurant tables, book event venues, customize catering packages, making secure online payments, and monitor the progress of their event reservations through an intuitive web interface. At the same time, restaurant administrators and staff will have access to tools for managing reservations, menus, inventory, employees, payments, vendors, and business reports.

To further enhance business decision-making, the system will incorporate data-driven analytical features such as dynamic pricing for event venues, intelligent food recommendations, customer sentiment analysis, demand forecasting, inventory optimization, and predictive staff allocation. These features will assist management in improving operational efficiency, reducing costs, and delivering a better customer experience.

The system will be developed as a **Java Spring Boot web application** following Software Engineering principles, including the Software Development Life Cycle (SDLC), Agile Scrum methodology, object-oriented design, UML modeling, and appropriate design patterns. The project will provide a scalable, secure, and user-friendly solution that addresses the operational needs of restaurant management and event planning.

3.2 Project Objectives

The primary objective of this project is to develop a modern web-based system that efficiently manages restaurant and event-related operations through a centralized platform.

The specific objectives of the project are:

- To provide customers with an online platform for restaurant table reservations and event venue bookings.
- To enable customers to customize catering packages based on their preferences and dietary requirements.

- To improve restaurant and event management by automating reservation, payment, inventory, and staff management processes.
- To reduce booking conflicts through real-time availability management.
- To provide secure online payment facilities and automated invoice generation.
- To support restaurant managers in making informed decisions through analytical dashboards and forecasting features.
- To improve customer satisfaction by offering personalized recommendations and efficient service management.
- To develop a scalable and maintainable web application using Java Spring Boot and modern software engineering practices.

3.3 Target Users/Stakeholders

The proposed system is intended to serve multiple stakeholders involved in restaurant and event management operations. Each stakeholder interacts with the system according to their specific responsibilities and requirements.

The primary stakeholders include:

- **Administrator** – Manages users, system settings, menus, reports, and overall system operations.
- **Customer** – Books restaurant tables and event venues, customizes catering menus, makes payments, and provides feedback.
- **Restaurant Manager** – Oversees restaurant operations, monitors reservations, manages pricing strategies, and analyzes business performance.
- **Event Coordinator** – Organizes event bookings, manages vendors, tracks event progress, and coordinates event-related services.
- **Chef** – Manages menu items, catering packages, recipes, and ingredient usage.
- **Cashier** – Processes customer payments, generates invoices, and manages billing transactions.
- **Inventory Manager** – Monitors stock levels, supplier information, ingredient usage, and inventory replenishment.
- **HR Manager** – Manages employee records, attendance, work schedules, and staff performance.
- **Restaurant Staff** – Performs daily operational task such as serving customers, updating order statuses, and supporting event activities.
- **Suppliers and Vendors** – Provide ingredients, event equipment, decorations, photography, entertainment, and other supporting services.

3.4 Project Scope

The scope of the proposed system includes the development of a complete web-based solution capable of managing restaurant dining services and event management activities within a single platform.

The system will provide functionality for:

- User registration and secure authentication.
- Restaurant table reservation and event venue booking.
- Interactive menu browsing and catering package customization.
- Vendor and event coordination management.
- Online payment processing and invoice generation.
- Inventory and stock management.
- Staff scheduling and attendance management.
- Customer feedback collection and reporting.
- Administrative dashboards and management reports.
- Data-driven analytical features to support business decision-making.

The project focuses on providing a responsive web application accessible through modern web browsers. Mobile applications and advanced third-party integrations will not be included in the initial implementation.

3.5 Project Limitations

The proposed system is developed within the scope and duration of a university Software Engineering project; therefore, several limitations are identified.

- The system will be developed as a web-based application and will not include native Android and iOS applications.
- Integration with real payment gateways may be simulated or implemented using sandbox environments for demonstration purposes.
- AI and data analytics features will be implemented using simplified predictive models suitable for academic purposes.
- The system assumes a stable internet connection for all online operations.
- Large-scale deployment features such as distributed cloud infrastructure and enterprise-level scalability are outside the project scope.
- Third-party vendor systems will not be directly integrated unless suitable public APIs are available.
- The project implementation will be completed within the semester timeline and available development resources.

4. System Overview

Web-Based Restaurant and Event Management System is a consolidated web-based system that targets managing the services provided by restaurants and events management on one platform. It provides an opportunity for communication between customers, administrators, restaurant workers, event organizers, and managers.

The customers can register into the system and use it to see the services offered by the restaurant, reserve tables, reserve event halls, customize catering services, and do payment transactions online. Real-time availability of tables and event halls would be available to help customers pick the right dates and times for booking purposes.

The system will be used by the administrators and managers to keep track of the menus, reservations, coordination of the vendors, inventory levels, scheduling of the staff, payment processing, and creation of operation reports. The staff members will update the progress of the events, assigned tasks, attendance, and activities through their respective dashboards.

A number of data analytics-based features will be made available to increase the efficiency of the operations and facilitate decision making for the company. These features are dynamic pricing according to the demand for the event spaces, intelligent menu recommendation according to the customer preference, sentiment analysis of the customer reviews, demand forecasting, and optimal inventory control to avoid wastage of food, and staff allocation.

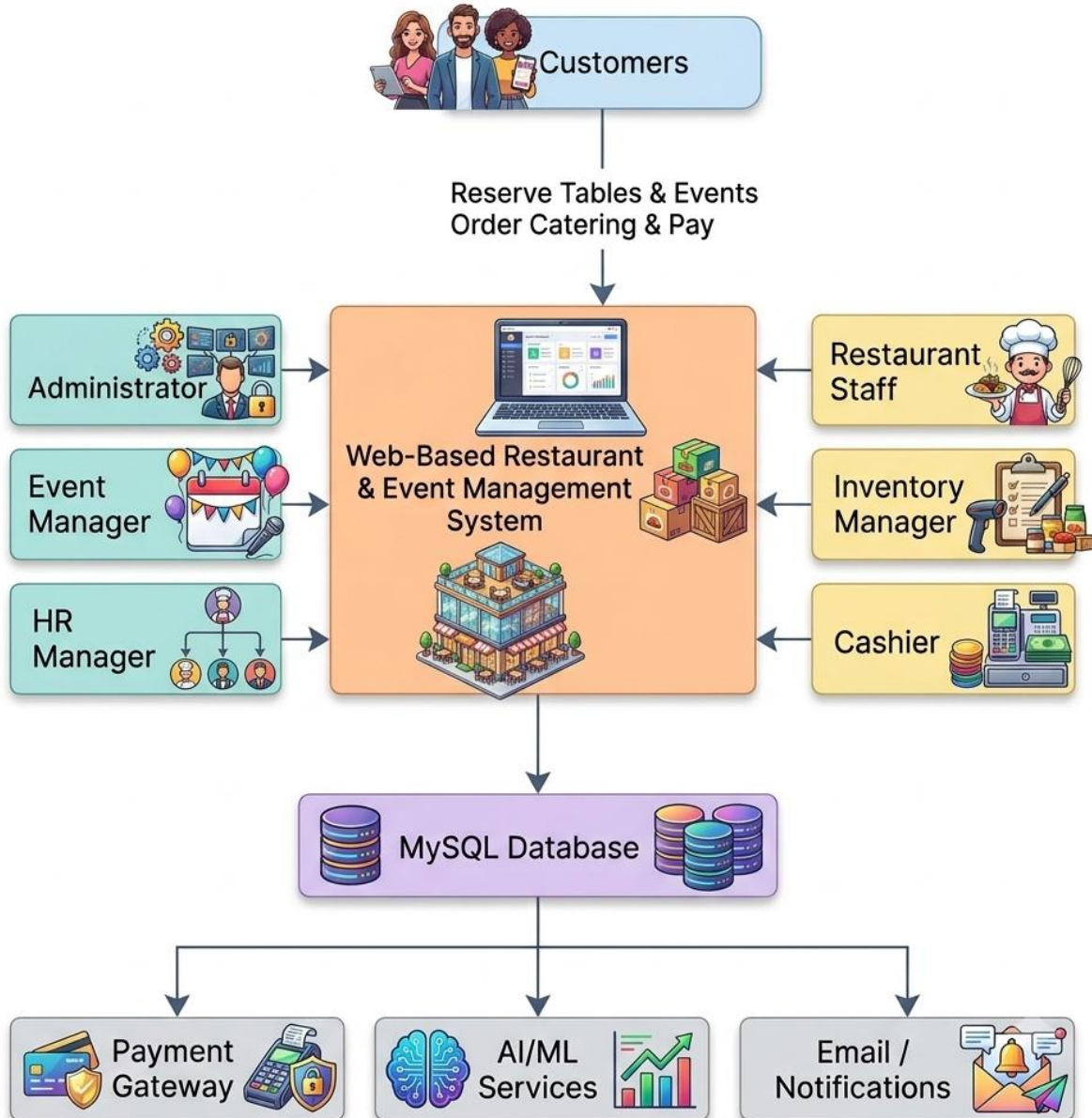
The system is implemented with the role-based access control mechanism, allowing the system users to log in according to their roles. Customers will be able to make reservations, employees will manage processes, managers will evaluate performance, and administrators will configure the system.

With respect to the technologies used in the development process, Java Spring Boot is used for the backend part of the system, while MySQL is used for the database and the modern web frontend solution.

This system is developed in order to accomplish the following purposes: improvement of service quality, satisfaction of customers, reduction of workload, effective management of resources, and provision of necessary business data to restaurant managers.

5. System Overview Diagram

The following diagram illustrates the overall architecture of the proposed **Web-Based Restaurant and Event Management System**. It shows the primary users of the system, the core system modules, and the external services that support the application's functionality.



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6. Functional Requirements

Functional requirements define the essential services and operations that the proposed **Web-Based Restaurant and Event Management System** must perform to satisfy the needs of its users. The system shall provide the following core functionalities.

Requirement ID	Functional Requirement	Description
FR-01	User Registration and Authentication	The system shall allow users to register, log in, log out, and securely manage their accounts based on role-based access control (Administrator, Customer, Staff, and Manager)
FR-02	Table Reservation Management	The system shall customers to view available tables select preferred seating, reserve tables, modify reservations, and cancel bookings while preventing double-booking through real-time availability checking.
FR-03	Event Venue Reservation	The system shall allow customers to reserve event halls, specify event details, select booking dates and times, and receive booking confirmations.
FR-04	Dynamic Venue Pricing	The system shall calculate event venue rental charges dynamically based on booking demand, weekends, holidays, and seasonal trends to optimize revenue.
FR-05	Menu Management and Customization	The system shall allow customers to browse menus, create customized catering packages, specify dietary preferences, and automatically calculate the total catering cost.

FR-06	Intelligent Food Recommendation	The system shall recommend suitable food items and beverages based on customer preferences, previous orders, dietary requirements, and popular menu combinations.
FR-07	Event Coordination Management	The system shall manage event - related services such as decorations, photography, entertainment, catering, and event schedules while tracking event progress through predefined milestones.
FR-08	Customer Feedback and Sentiment Analysis	The system shall allow customers to submit ratings and reviews after dining or events and analyze customer feedback to identify satisfaction trends.
FR-09	Billing and Payment Processing	The system shall generate invoices, process online payments, record advance deposits, support split-bill functionality, and maintain payment histories.
FR-10	Revenue Forecasting	The system shall analyze historical booking and sales information to forecast future demand and provide business insights for management.
FR-11	Inventory Management	The system shall management ingredients, beverages, equipment, suppliers, stock levels, and automatically deduct inventory quantities when customer orders or catering services are confirmed.
FR-12	Waste Monitoring and Stock Optimization	The system shall monitor food waste, generate low-stock alerts, and recommend appropriate reorder levels to improve inventory efficiency.
FR-13	Staff Management	The system shall maintain employee records, assign job roles, manage

		attendance, monitor working hours, and evaluate staff performance.
FR-14	Staff Scheduling	The system shall generate employee work schedules based on expected restaurant reservations and event bookings while allowing managers to adjust schedules when necessary.
FR-15	Notifications and Alerts	The system shall send reservation confirmations, payment receipts, booking reminders, schedule updates, and inventory alerts through email or system notifications.
FR-16	Reporting and Dashboard	The system shall provide dashboards and reports for administrators and managers, including status, customer feedback, and staff performance summaries.

7. Non-Functional Requirements

Non-functional requirements describe the qualities and standards that the Web-Based Restaurant and Event Management System must meet. These requirements focus on making sure the system is reliable, secure, easy to use and effective overall. They work alongside the requirements to explain how the system should behave in various situations.

Requirement ID	Non-Functional Requirement	Description
NFR-01	Performance	The system shall respond to user requests within 3 seconds under normal operating conditions. Realtime reservation availability and payment transactions should be processed efficiently with minimal delay.
NFR-02	Availability	The system shall be available 24 hours a day, 7 days a week , except during scheduled maintenance periods, ensuring uninterrupted access for customers and staff.
NFR-03	Security	The system shall provide secure user authentication, role-based access control (RBAC), encrypted password storage and secure communication to protect sensitive customer and business information.
NFR-04	Reliability	The system shall accurately process reservations, payments, inventory updates and staff schedules without data inconsistency or transaction failures.

NFR-05	Usability	The user interface shall be intuitive, responsive and easy to navigate for customers, administrator and staff with minimal training requirements.
NFR-06	Scalability	The system shall support future expansion, allowing additional restaurant branches, event venues, users and services to be incorporated with minimal system modifications.
NFR-07	Maintainability	The system be developed using a modular architecture, allowing developers to easily maintain, update and extend individual components without affecting the entire application.
NFR-08	Compatibility	The web application shall be compatible with modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firebox and Safari and shall support desktops, laptops and tablets.
NFR-09	Data Integrity	The System shall ensure that all reservation, payment, inventory and employee records remain accurate, consistent and protected from unauthorized modification.
NFR-10	Backup and Recovery	The system shall support regular database backups and recovery mechanisms to minimize data loss in the event of hardware or software failures.

NFR-11	Auditability	The system shall maintain activity logs for important operations such as user logins, reservations, payments, inventory updates and administrative actions to support monitoring and accountability.
NFR-12	portability	The application shall be deployable on standard web servers supporting Java Spring Boot and MySQL, allowing deployment in different organizational environments without significant modifications.

8. Major Stakeholders / Users

The suggested **Web-based Restaurant and Events Management System** caters to several stakeholders, each with different roles and levels of access. The system employs the **Role Based Access Control (RBAC)** mechanism to allow users access only to those functions which fall under their respective roles. The major stakeholders and their Roles are detailed below.

Administrator	Has complete control over the system. Manages users' accounts, restaurant data, menus, venues, generates reports, makes system settings and monitors overall performance of the system.
Customer	Registers an account, logs in to the system, reserves restaurant tables, reserves event venues, creates customized catering packages, does payment through the system, tracks reservation and submits reviews and feedback.
Restaurant Manager	Manages daily operations of the restaurant, monitors reservations, manages dining capacity, manages the staff, checks sales report and takes business decisions on the basis of system analysis.
Event Coordinator	Manages event booking, manages event scheduling, assigns service providers and tracks event preparations and communicates with customers for event arrangements.
Chef / Kitchen Staff	Manages food preparations, updates menus and monitors catering orders and ingredient requirements.

Cashier / Finance Officer	Performs billing tasks, handles customer payments, maintains deposit transactions, prepares bills, processes refund transactions, and maintains recordings of all payment transactions.
Inventory Manager	Maintains stock, coordinates with suppliers, maintains stock records, keeps track of ingredients used, receives notifications of low stock, and handles stock replenishments.
HR Manager	Manages employee information, assigns shifts to employees, manages attendance, evaluates employee performance, and coordinates workforce management.
Restaurant Staff	Refers to waiters, supervisors, and other service staff involved in customer service, managing status of reservations, handling restaurant operations, and helping out in events.
Suppliers / Vendors	Supply ingredients, beverages, decoration, photography, entertainment, sound system, and other event related material for the successful execution of events.

9. Six Major Functions (One per member)

The proposed Web-Based Restaurant and Event Management System consists of six major functional modules. Each module is assigned to one group member and focuses on a specific business area of the system. Together, these modules provide a complete solution for managing restaurant operations and event services.

9.1 Table & Venue Reservation with Dynamic Pricing

Assigned Member: IT25101990 – Amasha S.M.N

Who Will Use It

- Customers
- Restaurant Manager
- Reception Staff
- Administrator

What It Does

This module allows customers to reserve restaurant tables and event venues through an online booking system. Customers can view available tables or halls, choose their preferred seating or venue, and complete reservations based on real-time availability. The module also includes a dynamic pricing feature that adjusts event venue rental prices according to booking demand, weekends, holidays, and seasonal trends.

Key Sub-Functions

- View available tables and event halls
- Interactive table and venue selection
- Real-time availability checking
- Table and venue reservations
- Reservation modifications and cancellation
- Dynamic pricing calculations
- Reservation confirmation notifications

Expected Outcome

The module will provide a fast and reliable reservation process, eliminate double bookings, optimize venue utilization, and improve customer convenience while maximizing revenue intelligent pricing.

9.2 Menu Customization & Intelligent Recommendation

Assigned Member: IT25101976 – Undugodage J.C.A

Who Will Use It

- Customers
- Chef
- Restaurant Manager
- Administrator

What It Does

This module enables customers to browse restaurant menus and create customized catering packages for events. Customers can select dishes based on dietary preferences, serving sizes, and package types. The system also recommends complementary food items and beverages using customers preferences and ordering history.

Key Sub-Functions

- Browse menu categories
- Create custom catering packages
- Dietary and allergen filtering
- Automatic cost calculation
- Intelligent food recommendations
- Save favorite packages

Expected Outcome

The module will improve customer satisfaction by providing personalized menu options, simplify catering package creation, and increase restaurant sales through intelligent recommendations.

9.3 Event Coordination & Customer Feedback Analytics

Assigned Member: IT25102909 – Charuka M.G.D.D.V

Who Will Use It

- Event Coordinator
- Customer
- Vendor
- Restaurant Manager

What It Does

This module manages the complete event coordination process by organizing event schedules, assigning vendors, tracking event milestones, and monitoring service completion. After an event, customers can submit reviews and ratings, allowing the system to analyze customer satisfaction and identify service improvement opportunities.

Key Sub-Function

- Event booking management
- Vendor assignment
- Event timeline tracking
- Service progress monitoring
- Customer feedback collection
- Sentiment analysis dashboard

Expected Outcome

The module will improve event organization, enhance communication among stakeholders, and help management improve service quality through customer feedback analysis.

9.4 Billing, Payments & Revenue Forecasting

Assigned Member: IT25101082 – Perera G.N.N

Who Will Use It

- Customer
- Cashier
- Accountant
- Restaurant Manager

What It Does

This module manages customer payments for restaurant dining and event bookings. It supports secure online payments, invoice generation, split billing, and payment history management. The module also analyzes historical financial data to forecast future revenue and customer demand.

Key Sub-Functions

- Invoice generation
- Online payment processing
- Advanced deposit management
- Split bill functionality
- Payment history

- Revenue forecasting dashboard

Expected Outcome

The module will simplify financial operations, improve payment accuracy, reduce manual accounting tasks, and support management with financial planning.

9.5 Smart Inventory & Waste Reduction Management

Assigned Member: IT25103989 – Ahameth M.U.Y

Who Will Use It

- Inventory Manager
- Chef
- Restaurant Manager
- Administrator

What It Does

This module manages restaurant inventory by tracking ingredients, beverages, and event equipment. Inventory quantities are automatically updated when food orders or catering services are confirmed. The module also monitors food waste and recommends appropriate reorder levels to reduce operational costs.

Key Sub-Functions

- Inventory management
- Stock monitoring
- Automatic ingredient deduction
- Supplier management
- Low-stock alerts
- Waste tracking
- Safety stock calculation

Expected Outcome

The module will reduce inventory shortages, minimize food waste, improve stock accuracy, and support efficient inventory planning.

9.6 Staff Scheduling & Predictive Workforce Allocation

Assigned Member: IT25104028 – Niluminda B.G.E.S

Who Will Use It

- HR Manager
- Restaurant Manager
- Employees
- Administrator

What It Does

This module manages employee information, attendance, work schedules, and staff performance. Based on reservation volumes and event bookings, the system recommends appropriate staffing levels and assists managers in preparing efficient work schedules.

Key Sub-Functions

- Employee management
- Attendance recording
- Shift scheduling
- Leave management
- Performance tracking
- Predictive staff allocation
- Schedule notifications

Expected Outcome

The module will improve workforce management, ensure adequate staffing during busy periods, reduce scheduling conflicts, and enhance overall operational efficiency.

10. Minor Functions

In addition to the six main modules of functional components, the proposed Web-based Restaurant and Event Management System features some other functions which make the use of the system easier, more secure and more efficient. These functions are common for many modules and ensure efficient operation of the system.

User Registration	Able to enable new users to register by entering all personal details.
User Authentication	Enable login/logout functions based on Role-Based Access Control such that users can access relevant features.
Password Management	Enable the ability for the users to be able to modify passwords and be able to reset passwords through either email or generated link from the system.
User Profile Management	Enables users to view and update personal information, contact details, profile pictures, and account preferences
Notification Management	Enable the user to edit personal details, contact information, images/profile picture, and settings within their account. Send notification for confirmation, reservation, payment, schedule reminder, and alerts through email/web application.
Search and Filter	Enable searching and filtering of tables, venues, menu items, reservations, inventory, employees, and reports.
Dashboard Overview	Shows customized dashboards containing pertinent summaries, statistics, reservation reminders, events, payments, inventory notifications, and task assignments based on the user's position.
Report Generation	Creates operation and management reports, which include summaries of reservations, sales reports, inventory statuses, performance of staff, and customer feedback statistics.

Audit Trails	Logs significant user actions such as login details, reservations, payment transactions, inventory changes, and other administration activities.
Support and Help	Offers users system guidelines, FAQs, and technical support contact information.

11. System Limitations / Constraints

The Web-Based Restaurant and Event Management System being suggested here has been developed in order to meet all the requirements of restaurants and event management services. But owing to the limitations of the scope of the project and limited development period, certain limitations and constraints have been recognized.

Web-Based Platform Only	The system will be developed in the form of a web application and can be accessed using modern web browsers. Native applications for Android and iOS mobile platforms are not part of the initial development.
Internet Connectivity	Users must have a stable internet connection to access the system and perform online activities such as reservations, bookings, and payments.
Payment Gateway Integration	Online payment functionality may use a sandbox or simulated payment gateway during development and testing instead of a live payment service.
AI Feature Scope	Features like dynamic pricing, recommendation systems, demand forecasting, sentiment analysis, inventory optimization, and workforce planning based on AI will be developed using simpler models appropriate for the project context and available data
Browser Compatibility	The system is designed to be compatible with the latest browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari. Other browsers might not have all available features.
Single Business Environment	The first system version is meant for managing one restaurant and the corresponding services. The support for several branches can be considered in the future.
Third-Party Service Dependency	Features such as email notifications and online payment processing depend on external services. Temporary service interruptions may affect these functions

Project Time Constraints	The project will be developed within the academic semester timeline. Therefore, certain advanced features may be simplified while ensuring that all core functional requirements are successfully implemented
Data Accuracy Assumption	The system assumes that users and administrators enter accurate and complete information. Incorrect or incomplete data may affect reservation management, inventory records, reports, and system analytics.
Future Scalability	The proposed architecture will support future expansion; however, large-scale enterprise deployment, cloud-based distributed services, and high-volume transaction processing are beyond the scope of this project.

12. Week 3 – 14 Project Timeline

[illegible]

13. Conclusion

The proposed **Web-Based Restaurant and Event Management System** aims to provide a comprehensive and efficient solution for managing restaurant operations and event through a single integrated web platform. By combining essential business functions such as table and venue reservations, menu customization, event coordination, billing and payments, inventory management, and staff scheduling, the system seeks to improve operational efficiency while enhancing the overall customer experience.

In addition to these core functionalities, the project incorporates data-driven analytical features, including dynamic pricing, intelligent menu recommendations, customer sentiment analysis, demand forecasting, inventory optimization, and predictive staff allocation. These features will support informed decision making, improve resource utilization, and help restaurant management deliver high-quality services in a competitive business environment.

The project will be developed using **Java Spring Boot** and **MySQL**, following Software Engineering best practices, including the Software Development Life Cycle (SDLC), Agile Scrum methodology, object-oriented design principles, and UML-based system modeling. The modular architecture will ensure that the system is secure, maintainable, scalable and suitable for future enhancements.

The project responsibilities have been clearly distributed among the six group members, with each member responsible for developing one major functional module while collaborating on shared system components and integration. This structured approach promotes teamwork, effective project management, and balanced workload distribution throughout the development process.

The conclusion, the project presents a practical and innovative solution to modern restaurant and event management challenges. By replacing manual processes with an integrated digital platform, the proposed system has the potential to improve service quality, reduce operational complexity, and support better business decision-making. Our team is committed to following the proposed development timeline, maintaining effective communication, and delivering a reliable, high-quality system that satisfies the identified functional and non-functional requirements within the allocated project duration.